Neural-Network-Based Optimal Impedance Control for Robots in Physical Interaction With Soft Environments

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***Abstract*—With the growing demand for robots in emerging fields, such as smart medical and home services, their ability to interact with soft environments has received increased attention. Nevertheless, an overlooked issue is that the inadequate description of soft environments using a linear model may significantly diminish the accuracy of interaction control. In this article, a neural-network-based impedance control framework is proposed for robots to physically interact with soft environments and optimize interaction performance. Specifically, a nonlinear definition of soft environments is introduced based on the Hunt–Crossley (HC) model, with parameter identification utilizing a data-driven technique. Regarding system performance evaluated by a cost function, the determination of interaction behavior described by the impedance model is transformed into an optimal control problem. Moreover, to address model uncertainties, the original optimal control problem is redefined using a modified cost function with a constructed auxiliary system. Then, a critic network is employed to approximate the nonlinear optimal solution, thereby avoiding complicated mathematical derivations. Finally, the effectiveness of the proposed impedance adaptation strategy is validated through both simulations and experiments. Numerical results indicate that both the convergent cost and total cost are significantly reduced based on the proposed method compared to the linear-model-based impedance control, particularly for materials with viscoelastic properties, achieving a reduction of up to 30%.**

***Index Terms*—Hunt–Crossley (HC) model, neural network (NN), optimal impedance control, soft environments.**

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I. Introduction

IMPEDANCE control, characterized by regulating the relation between contact position and force, has been extensively applied for robotic contact operations, such as polishing and assembly in manufacturing [1], [2], [3], and body scanning and surgery in medical applications [4], [5]. Recent developments have further enhanced its versatility by integrating it with advanced control strategies. Huang et al. [6] incorporated the impedance control into the model predictive control (MPC) framework for real-time interactive robotic tasks. By leveraging the ability of MPC to predict future states and impedance control to adjust compliance, this strategy is capable of handling multiple unknown contacts. Liang et al. [7] combined force control with impedance control for an ultrasonic end effector interacting with soft environments. Logothetis et al. [8] proposed a variable impedance strategy for object manipulation. By changing the damping and stiffness parameters based on the measured slippage velocity, a stable grasp can be ensured. Furthermore, learning-based control methods, particularly reinforcement learning, enable robots to autonomously adjust impedance properties based on experiential data. Wang et al. [9] utilized Q-function-based critic learning method to adapt the impedance model to a time-varying environment, achieving optimal interaction performance. Subsequent studies addressed more complex application scenarios. For example, Liu et al. [10] investigated the optimal interaction between manipulators and environments with varying surface contours. The desired impedance model is learned by the output-feedback adaptive dynamic programming (OPFB ADP) method. Ye et al. [11] proposed an optimal impedance–admittance switching scheme using an iterative linear quadratic regulator (ILQR) to manage interaction issues in environments with varying stiffness. More detailed information regarding the characteristics of the environment, strengths, and weaknesses of the aforementioned impedance-based approaches can be found in Table I.

In the aforementioned studies, the environment is generally assumed to be representable by a linear model. The most commonly used linear models for describing environments in real-time interaction control are primarily limited to the Kelvin–Voigt (KV) model or the mass–spring–damper system [12], [13], [14]. Studies [15], [16], [17] demonstrated the effectiveness of these linear models in stiff environments. However, in soft-environment operations, physical discontinuities and negative characteristics of the contact force arise

TABLE I
COMPARISON OF THE MENTIONED IMPEDANCE-BASED CONTROL APPROACHES

Approach	Environmental Information	Interactive Dynamics	Impedance Model	Strength	Weakness
MPC framework [6]	online learning with prior model	multi-contact dynamics	fixed	real-time adaptability; effective for unknown non-smooth-surface objects	high computation; performance affected by prior model
Force tracking [7]	prior knowledge	Hunt-Crossley model	fixed	robust force tracking; accurate modeling of soft objects	relies on precise Hunt-Crossley model parameters
Variable impedance [8]	prior knowledge	multi-contact dynamics	variable	mitigates slippage effects;trajectory tracking convergence	dependent on slippage sensors
Q-function-based adaptive impedance [9]	unknown	time-varying linear system	adaptive	handles time-varying dynamics without full system knowledge	idealized assumptions limit deployment
OPFB-ADP-based adaptive impedance [10]	unknown	mass-damping-stiffness system	adaptive	achieves trajectory tracking with an arbitrary endpoint; faster convergence	limits to varying positions of contact objects representable by linear systems
ILQR-based adaptive impedance [11]	unknown	mass-damping-stiffness system	adaptive	combines impedance and admittance for adaptability	implementation complexity; requires careful tuning to avoid instability in switching

due to the viscoelastic properties of soft materials [18]. These issues indicate that linear models are insufficient for accurately describing soft environments, potentially degrading control performance [19]. To address the limitations of linear models, Hunt and Crossley proposed a nonlinear model, known as the Hunt–Crossley (HC) model, in which the damping coefficient is dependent on contact penetration [20]. The HC model can characterize the dynamics of both stiff and soft environments, and it has therefore been increasingly adopted for the robotic interaction control [21]. Different from these existing studies, this article focuses on the nonlinear model and optimal interaction performance suitable for soft-environment interactions.

The HC model retains a certain computational simplicity despite it being nonlinear [22]. Typical HC model identification utilizes position and contact force data collected during the robot's movements along an excitation trajectory [23]. Since the computational complexity of the identification methods significantly affects real-time robotic contact control, much research has focused on reducing the convergence time of model estimation to enhance control performance. The single-stage linearized logarithm identification (LLI) method proposed by Haddadi and Hashtrudi-Zaad [24] effectively shortens convergence time and is insensitive to initial parameters, making it popularly used for HC model identification. Schindeler and Hashtrudi-Zaad [25] focused on highly damped environment situations and produced a real-time identification method involving a linearly parameterized polynomial approximation to indirectly identify the HC model. Subsequently, they extend the identification into a two-stage strategy to achieve a more accurate estimation of the HC model parameters [23]. However, data-driven identification methods inevitably introduce noise into the system. Model uncertainties, including modeling error and force measurement noise, should be considered in a complete HC model [26]. In practice, model uncertainties are generally unknown, but the exact upper bound can be determined based on the ranges of the parameter variations in specific problems [27].

For optimal interaction, the interaction dynamics and performance index need to be developed first. Then, the impedance model is updated by solving the optimal control problem of the interaction system [28]. From a mathematical point of view, the solution to the optimal control problem is derived by solving the underlying Hamilton–Jacobi–Bellman (HJB) equation [29]. However, it is generally difficult to obtain the analytical solution for a nonlinear system. The adaptive dynamic programming (ADP) method offers an intuitive approach to solving the optimal control problem. It can be implemented using neural networks (NNs), which possess powerful approximation capabilities to adaptively learn the solution of the HJB equation [30]. A typical structure in ADP is the actor–critic dual setup, where the actor and critic NNs parameterize the control policy and cost function, respectively [31]. The optimal solution is reached after the two NNs iteratively train each other successfully [32]. A single-critic network structure, which removes the action network from the architecture, is considered an improvement over the actor–critic structure [33]. It retains the power characteristics of the actor–critic structure while reducing the computational load and eliminating the approximation error caused by the action network, making it more effective for online implementation. Kong et al. [34] employed the critic-based learning algorithm to address the tracking control problem for a highly nonlinear and strongly coupled robot system subject to unknown perturbations. Luo et al. [35] proposed a decentralized event-triggered control method for interconnected nonlinear systems with unknown perturbations using the critic networks technique. To our knowledge, few interaction control strategies simultaneously address the nonlinear environmental dynamics and model uncertainties, which motivates our work.

In this article, we investigate optimal interaction control between robots and soft environments. Position-based impedance control, also known as admittance control [36], is employed to achieve compliant robotic behavior. Considering that interaction performance heavily depends on the environmental dynamics and the impedance model, we introduce a nonlinear HC model to describe the soft environment, while the target impedance model is selected by minimizing a combined generalized force and position error metric. Specifically, the HC model is identified using a data-driven technique, after which the interaction dynamics is established accordingly. Moreover, to address model uncertainties in practical application, an auxiliary system is constructed, and the interaction performance is evaluated using a modified cost function. Based on the optimal control technique, the target impedance model is obtained by solving the optimal control problem. Given the nonlinear nature of the environmental model, a single-critic network is introduced to iteratively learn the control policy. In addition, the weight estimation error is guaranteed to be uniformly ultimately bounded (UUB) by incorporating the persistence of excitation (PE) assumption. The main contributions are highlighted as follows.

1) A comprehensive representation of the environmental dynamics, based on a nonlinear HC model and incorporating a disturbance term, is developed for robotic interaction with soft environments. These enhancements improve the modeling accuracy in representing real-world scenarios, particularly when dealing with viscoelastic materials that are inaccurate in linear approximations.
2) Model uncertainties, including modeling errors and force measurement noise resulting from data-driven model identification, are considered in the impedance adaptation strategy. In this context, an auxiliary control is incorporated into the system to transform the original robust control problem into a generally optimal control problem.
3) The desired impedance model to achieve a predefined metric is obtained by solving the optimal control problem of the nonlinear interactive system, where a single-critic network is employed to learn the optimal solution online.

The remainder of this article is organized as follows. In Section II, we present the preliminaries, including important system models, robust control strategy, and problem statement. The detailed of the impedance control design is explained in Section III. The simulations and experimental studies are provided in Sections IV and V. Section VI draws the conclusion.

II. Preliminaries

A. System Modeling and Identification

The contact dynamics of the environment can be modeled by the HC model [20] in the form of

$$f = B_e(x - x_e)^n\dot{x} + K_e(x - x_e)^n \tag{1}$$

where f denotes the interaction force, $\dot{x}$ and x represent the velocity and position of the end effector of the manipulator, respectively, and x_e is the equilibrium position of the environment. n is a constant, which is determined by the material and geometric properties of the environment. B_e and K_e are the damping and stiffness parameters related to the restitution coefficient, respectively.

Remark 1: The HC model (1) can represent various soft environments. For example, it can describe the reaction dynamics of rubber that demonstrates elastic behavior under contact forces [12], and typically has a Young's modulus in the range of 0.01–0.1 GPa. It is also suitable for deformable synthetic materials, such as polyurethane foam [23], which exhibit nonlinear compression and have Young's modulus values around 10–100 kPa. Additionally, the model can effectively capture the viscoelastic response of biological tissues (e.g., skin or muscle) [18], which cannot be accurately approximated using linear models, with typical stiffness ranging from 0.1 to 100 kPa.

Remark 2: The environmental equilibrium position x_e can be confirmed by incorporating a depth camera into the system. Upon getting the depth image of the environment, three-dimensional (3-D) point cloud technique can be employed to represent the environment into data points [37]. Then, x_e is obtained via taking the minimum Euclidean distance of the current position with the data points.

To identify the parameters B_e, K_e, and n, the manipulator is commanded to follow an excitation trajectory, and the data of force, position, and velocity is collected. Then, a single-stage linearized logarithmic identification method [24] is employed. Considering the noises in practical operation, it is necessary to add a term of model uncertainties ϵ to ensure the integrity of the environmental dynamics. Taking the natural logarithm of (1)

$$\ln(f) = \ln\left(K_e(x - x_e)^n(1 + \frac{B_e\dot{x}}{K_e} + \frac{\epsilon}{K_e(x - x_e)})\right). \tag{2}$$

Since $\ln(1 + \varpi) \cong \varpi$ for $|\varpi| \ll 1$, it can be assumed that

$$|\frac{B_e\dot{x}}{K_e} + \frac{\epsilon}{K_e(x - x_e)}| \ll 1. \tag{3}$$

Then, (2) can be linearized as

$$\ln(f) \cong \ln(K_e) + n\ln(x - x_e) + \frac{B_e\dot{x}}{K_e} + \frac{\epsilon}{K_e(x - x_e)}. \tag{4}$$

Remark 3: The velocity of the end effector of the manipulator can be restricted to $|\dot{x}|_\infty < (0.1K_e/B_e)$. Hence, it is sufficient to meet $|(B_e\dot{x}/K_e)| \ll 1$. On the other hand, ϵ can be regarded as a zero-mean stochastic process, which satisfies the mathematical expectation of $(\epsilon/[K_e(x - x_e)])$ is zero. Therefore, $|([B_e\dot{x}]/K_e) + (\epsilon/[K_e(x - x_e)])| \le |([B_e\dot{x}]/K_e)| + |(\epsilon/[K_e(x - x_e)])| \ll 1$.

Further, (4) can be parameterized as

$$y_k = \Theta_k^T\varphi + \tilde{\epsilon}_k \tag{5}$$

where k is the discrete-time index, $y_k = \ln(f_k)$, $\Theta_k^T = [1, \dot{x}_k, \ln(x_k - x_e)]$, $\varphi = [\ln(K_e), (B_e/K_e), n]$. $\tilde{\epsilon}_k = (\epsilon_k/[K_e(x_k - x_e)]) + \bar{\epsilon}_k$ denotes the noises and modeling error.

To estimate the parameter φ, the exponentially weighted recursive least squares (EWRLS) method [38] is employed

$$\Lambda_{k+1} = \frac{\Omega_k\Theta_{k+1}}{\gamma + \Theta_{k+1}^T\Omega_k\Theta_{k+1}} \tag{6}$$

$$\Omega_{k+1} = \frac{1}{\gamma}\left[\Omega_k - \Lambda_{k+1}\Theta_{k+1}^T\Omega_k\right] \tag{7}$$

$$\hat{\varphi}_{k+1} = \hat{\varphi}_k + \Lambda_{k+1}\left[y_{k+1} - \Theta_{k+1}^T\hat{\varphi}_k\right] \tag{8}$$

where γ is a forgetting factor. Λ represents the adaption gain, and Ω denotes the covariance matrix. Therefore, the environmental parameters can be derived from

$$\hat{K}_{ek} = e^{\hat{\varphi}_k(1)} \quad \hat{B}_{ek} = e^{\hat{\varphi}_k(1)}\hat{\varphi}_k(2) \quad \hat{n}_k = \hat{\varphi}_k(3) \tag{9}$$

where $\hat{\varphi}(i)$ is the ith element of vector $\hat{\varphi}$.

B. Robust Control for Nonlinear Systems

The nonlinear system with model uncertainties can be expressed as

$$\dot{x} = A(x) + B(x)u + k(x)w(x) \tag{10}$$

where $x \in \mathcal{R}^{\varsigma}$ is the system state, $u \in \mathcal{R}^{\iota}$ is the control input, $A(\cdot) : \mathcal{R}^{\varsigma} \to \mathcal{R}^{\varsigma}$ is the internal state dynamics, $B(\cdot) : \mathcal{R}^{\varsigma} \to \mathcal{R}^{\varsigma\times\iota}$ is the input-to-state dynamics, and $k(x)w(x)$ represents unknown model uncertainties with $w(\cdot) : \mathcal{R}^{\varsigma} \to \mathcal{R}^{\tau}$, $k(\cdot) : \mathcal{R}^{\varsigma} \to \mathcal{R}^{\varsigma\times\tau}$. Without loss of generality, we assume that $A(x_s) = 0$ and $w(x_s) = 0$ can always be met at a certain state x_s so that x_s is an equilibrium position of the system. Using the orthogonal decomposition property, model uncertainties $k(x)w(x)$ can always be decomposed into the sum of an unmatched component and a matched component. Thus, we have

$$\begin{aligned}\dot{x} = A(x) + B(x)u &+ \left[I_{\varsigma} - B(x)B(x)^*\right]k(x)w(x) \\ &+ B(x)B(x)^*k(x)w(x)\end{aligned} \tag{11}$$

where I_{ς} represents the unit matrix in the dimension of ς. $B(x)B(x)^*k(x)w(x)$ is the matched component, and $[I_{\varsigma} - B(x)B(x)^*]k(x)w(x)$ is the unmatched component. $(\cdot)^*$ represents the pseudo-inverse operation. Accordingly, we can build an auxiliary system

$$\dot{x} = A(x) + B(x)u + \left[I_{\varsigma} - B(x)B(x)^*\right]k(x)v(x) \tag{12}$$

where v is the auxiliary control input.

Assume there exists non-negative functions $w_{\max}(x)$ and $\zeta_{\max}(x)$ that satisfy $\|w(x)\| \leq w_{\max}(x)$ and $\|B(x)^*k(x)w(x)\| \leq \zeta_{\max}(x)$, respectively. Then, a cost function can be designed as

$$\int_0^{\infty} \zeta_{\max}(x)^2 + \varrho^2 w_{\max}(x)^2 + \delta^2\|x\|^2 + \|u\|^2 + \varrho^2\|v\|^2 \, d\tau \tag{13}$$

where ϱ and δ are designed parameters.

Theorem 1 [39]: If one can find a solution $(u_0(x), v_0(x))$ for the optimal control problem that minimizes the cost function (13) and satisfies

$$\varrho^2\|v_0(x)\|^2 \leq \delta'^2\|x\|^2 \quad \forall x \in \mathcal{R}^{\varsigma} \tag{14}$$

for some $\delta' < \delta$, then $u_0(x)$ is also a robust controller for the nonlinear system (10) so that the system state is globally asymptotically stable for the uncertainties $w(x)$.

In particular, when the uncertainties without the unmatched component, that is

$$\left[I_{\varsigma} - B(x)B(x)^*\right]k(x)w(x) = 0 \tag{15}$$

the robust control problem reduces to the optimal control problem for the auxiliary system $\dot{x} = A(x) + B(x)u$. The cost function can be simplified to

$$\int_0^{\infty} \zeta_{\max}(x)^2 + \|x\|^2 + \|u\|^2 \, d\tau \tag{16}$$

by setting $\varrho = 0$ and $\delta = 1$.

C. Impedance Control

Impedance control allows for compromise between force and position error, which can be expressed as

$$f = \mathcal{I}(x - x_d). \tag{17}$$

f is the interaction force, x is the system state, and x_d is the desired position. $\mathcal{I}(\cdot)$ represents the target impedance, which can be in the form of a linear damping–spring system

$$\mathcal{I}(s) = B_a s + K_a \tag{18}$$

where s is the Laplace variable, and B_a and K_a are the damping and stiffness parameters, respectively.

Remark 4: The choice of a specific target impedance for the optimal interaction depends on the interactive environment. Generally, for a damping–stiffness environment, only stiffness control is required, whereas a damping component needs to be added for a mass–damping–stiffness environment [40].

Target impedance can be realized via position-based implementation. The control command f_c is derived by

$$f_c = \mathcal{C}(s)(x_r - x) \tag{19}$$

where $\mathcal{C}(s)$ is the inner-loop position controller, the reference position x_r is a perturbation from x_d and has

$$x_r = x_d + \frac{f}{\mathcal{I}(s)}. \tag{20}$$

We assume that $\mathcal{C}(s)$ is accurate enough, so that $\lim_{t\to\infty} x(t) = x_r(t)$ for realizing target dynamics (17).

D. Problem Formulation

We consider a single-point interaction of the manipulator end effector along one direction with a soft environment. The desired position x_d is given by a command generator

$$\begin{cases}\dot{r} = \Gamma r + \alpha \\ x_d = \Xi r\end{cases} \tag{21}$$

where $r \in \mathcal{R}^n$ is the state, and $\Gamma \in \mathcal{R}^{n\times n}$, $\alpha \in \mathcal{R}^n$, and $\Xi \in \mathcal{R}^{1\times n}$ are constant parameters.

For a general tracking control problem, the control objective is to enable the robot to precisely track the desired trajectory. However, in the robotic control system for interaction with environments, contact inevitably occurs and may cause physical damage. On the other hand, although impedance control can effectively regulate the force in interactions and endow the robot with compliant behaviors to safely interact with environments, it cannot satisfy the tracking control accuracy. Therefore, both the position tracking error and force should be considered at the same time in the control design. To cover these two indexes, the cost function to evaluate the system performance is designed as

$$\bar{J}(X(t)) = \int_t^\infty e^T Q e + f^T R f \, d\tau \tag{22}$$

where $e := x - x_d$ represents the position tracking error. Q and R are the weight indexes. From (22), it can be noted that the performance of the system describes a tradeoff between the position tracking error and the interaction force. As a result, the control objective is to design a control policy to minimize the cost function (22), which is actually solving a quadratic regulation problem [41].

Remark 5: In practical implementations, the different units of measurement result in different quantities of weight indexes to achieve the desired system performance. Partial knowledge of both the robot and the environment is often leveraged to select their initial values. Subsequently, a trial-and-error approach can be employed to refine the weights based on system behavior.

Remark 6: The cost function (22) is designed in the demand of balancing position tracking accuracy $e^T Q e$ and the energy efficiency $f^T R f$, with the desired force considered to be zero. In applications, such as robotic surgery, material handling, or human–robot collaboration, reducing excessive forces while maintaining precise position tracking is critical to avoiding damage to the environment or the robot itself [42].

Remark 7: The total cost $\bar{J}(X(t))$, as defined in (22), quantifies the cumulative interaction effort over the entire process, reflecting overall control efficiency and smooth adaptation. In contrast, the convergent cost $\lim_{t\to\infty} \bar{J}(X(t))$ measures the instantaneous cost after the system reaches steady state, capturing the quality of final regulation and long-term interaction stability. These two key metrics are sufficient to provide a comprehensive evaluation of the interaction performance.

In the next section, we will provide a detailed introduction to how to achieve the objective by designing a robot impedance controller and implementing a position-based scheme.

III. Controller Design

In this section, the detailed control design for the optimal interaction with a soft environment is given. First, we construct an augmented system and transfer the adaption problem of the impedance model into an optimal control problem. Next, the HJB equation of the optimization problem and the optimal control policy are derived. Then, to avoid the complex mathematical calculation for the HJB solution, an NN-based critic learning strategy is proposed and the convergence analysis of the NN is provided. Finally, the desired impedance model is obtained and realized through position-based implementation. The whole system framework is shown in Fig. 1.

A. Impedance Adaptation

The HC model can be rewritten into a state-space form

$$\dot{x} = -\frac{K_e}{B_e} + \frac{f}{B_e(x - x_e)^n} + \varepsilon \tag{23}$$

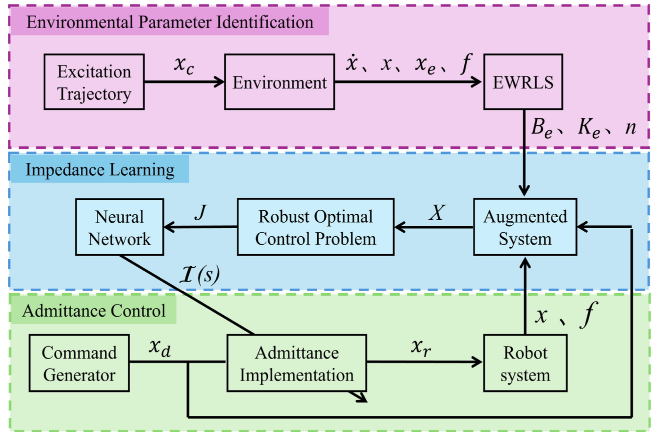

Fig. 1. Proposed control framework for optimal robot–environment interaction.

where $\varepsilon = -(\epsilon/[B_e(x - x_e)^n])$ stands for the unknown model uncertainty term introduced for integrality of the environmental dynamics.

Putting the dynamics of the environment position and the desired position together, an augmented system can be constructed

$$\dot{X}(t) = F(X(t)) + G(X(t))f + K\varepsilon(X(t)) \tag{24}$$

where $X(t) = \begin{bmatrix} e \\ r \end{bmatrix}$, $F(X(t)) = \begin{bmatrix} -(K_e/B_e) - \Xi\Gamma r - \Xi\alpha \\ \Gamma r + \alpha \end{bmatrix}$, $G(X(t)) = \begin{bmatrix} (1/[B_e(x - x_e)^n]) \\ O_n \end{bmatrix}$, and $K = \begin{bmatrix} 1 \\ O_n \end{bmatrix}$, O_n is the zero vector in the dimension of n, and f is regarded as the control input.

Considering that the model uncertainties in the augmented system (24) only contain the matched component

$$[I_{1+n} - GG^*]K\varepsilon = O_{1+n}$$

an auxiliary system is introduced

$$\dot{X}(t) = F(X(t)) + G(X(t))f. \tag{25}$$

Accordingly, a modified cost function J for the auxiliary system is designed as

$$\begin{aligned} J(X(t)) &= \int_t^\infty e^T Q e + f^T R f + \zeta_\varepsilon^2(X) \, d\tau \\ &= \int_t^\infty X^T \tilde{Q} X + f^T R f + \zeta_\varepsilon^2(X) \, d\tau \end{aligned} \tag{26}$$

where $\tilde{Q} = \begin{bmatrix} Q & O_n^T \\ O_n & O_{n\times n} \end{bmatrix}$, $O_{n\times n}$ is a zero matrix in the dimension of n, $\zeta_\varepsilon(X)$ is chosen to satisfy $\|\varepsilon\| \le \zeta_\varepsilon(X)$. The analytical solution for the optimal control problem (25)-(26) is an expression describing the relation between f and X, which is exactly the required impedance model.

Remark 8: The auxiliary system is employed for compensating the uncertainties in the interactive dynamics. According to Theorem 1, the optimal control for the auxiliary system (25) with cost function (26) can be considered as a robust optimal control of a nonlinear system with model uncertainties (24). Since the environmental uncertainties only contain the matched component, the cost function can be designed as consisting of three terms, where $X^T\tilde{Q}X$ and f^TRf are the costs for standard optimal control, and $\zeta_\varepsilon^2(X)$ for the cost of uncertainties compensation. In this way, a desired control policy not only achieving optimal interaction performance but also stabilizing the system under model uncertainties can be obtained.

B. HJB Equation

Assuming that the cost function (26) is continuous and differentiable, and the considered optimal control policy f^* is within the set of admissible control, i.e., $f \in \Psi$. Rewrite (26) in the infinitesimal version

$$(\nabla J)^T(F+Gf)+X^T\tilde{Q}X+f^TRf+\zeta_\varepsilon^2=0 \tag{27}$$

with $J(0)=0$, where $\nabla J=\partial J/\partial X$.

Define the Hamiltonian

$$\mathcal{H}(X,f,\nabla J)\ =(\nabla J)^T(F+Gf)+X^T\tilde{Q}X+f^TRf+\zeta_\varepsilon^2. \tag{28}$$

According to Bellman's optimality principle, we have the HJB equation

$$\min_{f\in\Psi}\ \mathcal{H}\big(X,f,\nabla J^*\big)=0. \tag{29}$$

Supposing that the optimal control policy exists and is unique, we have

$$\begin{aligned} f^* &= \arg\min_{f\in\Psi}\ \mathcal{H}\big(X,f,\nabla J^*\big)\\ &= -\frac{1}{2}R^{-1}G^T\nabla J^*. \end{aligned} \tag{30}$$

By substituting (30) into (29) and solving this HJB equation, the optimal cost function J^* can be acquired, so the optimal control policy is also available based on (30). However, the analytic solution of the nonlinear HJB equation (29) is generally difficult to derive.

C. NN-Based Approximate of HJB Solution

Utilizing the approximation ability of NN, we can design a critic learning strategy to obtain the approximate HJB solution. The optimal cost function J^* can be represented by

$$J^*(X)=W^T\phi(X)+\sigma(X) \tag{31}$$

where $\phi(X):\mathcal{R}^{1+n}\to\mathcal{R}^m$ represents the activation function, $W\in\mathcal{R}^m$ is the ideal weight vector, m denotes the number of hidden layer neurons, and $\sigma(X)$ is the approximation error. Accordingly, its derivative can be represented by

$$\begin{aligned} \nabla J^*(X) &= \left(\frac{\partial\phi(X)}{\partial X}\right)^T W+\frac{\partial\sigma(X)}{\partial X}\\ &= \nabla\phi(X)^TW+\nabla\sigma(X) \end{aligned} \tag{32}$$

where $\nabla\phi\in\mathcal{R}^{m\times(1+n)}$, $\nabla\sigma\in\mathcal{R}^{1+n}$. For constant m, σ and $\nabla\sigma$ are bounded by constants on a compact set [31]. When $m\to\infty$, $\sigma\to 0$, so $\nabla\sigma\to 0$. In this way, the optimal control policy can be obtained by

$$f^*(X)=-\frac{1}{2}R^{-1}G^T\big(\nabla\phi(X)^TW+\nabla\sigma(X)\big) \tag{33}$$

and the HJB equation satisfies

$$\mathcal{H}\big(X,f^*,W\big)=0. \tag{34}$$

However, the ideal weight W is unknown in practice. The approximate result of the cost function is

$$\hat{J}(X)=\hat{W}^T\phi(X) \tag{35}$$

where $\hat{W}$ is the weight vector for the current estimate. Taking the derivative of (35) yields

$$\nabla\hat{J}(X)=\nabla\phi(X)^T\hat{W}. \tag{36}$$

Similarly, we can get the approximate control policy

$$\hat{f}(X)=-\frac{1}{2}R^{-1}G^T\nabla\phi(X)^T\hat{W}. \tag{37}$$

Then, the approximate Hamiltonian can be expressed as

$$\hat{\mathcal{H}}\Big(X,\hat{f},\hat{W}\Big)=\hat{W}^T\nabla\phi\Big(F+G\hat{f}\Big)+X^T\tilde{Q}X+\hat{f}^TR\hat{f}+\zeta_\varepsilon^2. \tag{38}$$

Define the approximation error of Hamiltonian $\sigma_H := \hat{\mathcal{H}}(X,\hat{f},\hat{W})-\mathcal{H}(X,f^*,W)$. Then, we have

$$\sigma_H=\hat{W}^T\nabla\phi\Big(F+G\hat{f}\Big)+X^T\tilde{Q}X+\hat{f}^TR\hat{f}+\zeta_\varepsilon^2. \tag{39}$$

The NN weight is chosen to minimize

$$E=\frac{1}{2}\sigma_H^T\sigma_H \tag{40}$$

so that $\sigma_H\to 0$, $\hat{W}\to W$.

The updating law for the NN weight is designed as

$$\begin{aligned} \dot{\hat{W}} &= -\frac{\kappa}{\left(1+\nu^T\nu\right)^2}\left(\frac{\partial E}{\partial\hat{W}}\right)\\ &= -\frac{\kappa\nu}{\left(1+\nu^T\nu\right)^2}\Big[\nu^T\hat{W}+X^T\tilde{Q}X+\hat{f}^TR\hat{f}+\zeta_\varepsilon^2\Big] \end{aligned} \tag{41}$$

where $\nu=\nabla\phi(F+G\hat{f})$. κ is the learning factor. Define the weight estimation error $\tilde{W}:=W-\hat{W}$. Then, the differential of the weight estimation error is

$$\dot{\tilde{W}}=-\dot{\hat{W}}=\frac{\kappa\nu}{\left(1+\nu^T\nu\right)^2}\big(\sigma_a-\nu^T\tilde{W}\big) \tag{42}$$

where $\sigma_a=\nabla\sigma^T(F+G\hat{f})$ represents the residual error and is assumed to be bounded $\sigma_a\le\eta_a$ [35]. To guarantee the convergence of $\hat{W}$ to W, we design a Lyapunov function candidate

$$L=\frac{1}{2}\tilde{W}^T\tilde{W}. \tag{43}$$

Differentiating L with respect to time yields

$$\begin{aligned} \dot{L} &= -\frac{\kappa\tilde{W}^T\nu}{\left(1+\nu^T\nu\right)^2}\big(\nu^T\tilde{W}-\sigma_a\big)\\ &\le -\frac{\kappa\tilde{W}^T\nu\nu^T\tilde{W}}{\left(1+\nu^T\nu\right)^2}+\kappa\frac{\tilde{W}^T\nu\nu^T\tilde{W}+\sigma_a^T\sigma_a}{2\left(1+\nu^T\nu\right)^2}. \end{aligned} \tag{44}$$

Note that $\sigma_a \leq \eta_a$ and $1 + \nu^T \nu \geq 1$, thus we have

$$\begin{aligned}\dot{L} &\leq -\frac{\kappa \tilde{W}^T \nu \nu^T \tilde{W}}{2\left(1+\nu^T \nu\right)^2} + \frac{\kappa}{2}\eta_a^2 \\ &\leq -\frac{\kappa}{2}\lambda_{\min}\left(\xi\xi^T\right)\|\tilde{W}\|^2 + \frac{\kappa}{2}\eta_a^2 \end{aligned} \tag{45}$$

where $\xi = (\nu/[1+\nu^T\nu])$, $\lambda_{\min}(\xi\xi^T)$ stands for the minimum eigenvalue of $\xi^T\xi$.

It is obvious that if the inequalities

$$\lambda_{\min}\left(\xi\xi^T\right) > 0 \tag{46}$$

$$\|\tilde{W}\| > \frac{\eta_a}{\sqrt{\lambda_{\min}\left(\xi\xi^T\right)}} \triangleq \Upsilon \tag{47}$$

hold, $\dot{L} < 0$ can be guaranteed. To satisfy $\lambda_{\min}(\xi\xi^T) > 0$, we introduce the PE assumption: the signal ξ is persistently exciting over the interval $[t, t+T]$ with $T > 0$, i.e., the condition

$$\upsilon_1 I_m \leq \int_t^{t+T} \xi(\tau)\xi^T(\tau)\, d\tau \leq \upsilon_2 I_m \tag{48}$$

is met for all t, where $\upsilon_1 > 0$ and $\upsilon_2 > 0$ are constants. Therefore, the weight estimation error $\tilde{W}$ is UUB and the ultimate bound is Υ.

D. Reference Generation via NN-Based Admittance Model

Regarding the optimal control policy implemented by NN (37), we can expand it to yield

$$\hat{f} = -\frac{1}{2R}\begin{bmatrix} G_1 \\ \vdots \\ G_N \end{bmatrix}^T \begin{bmatrix} \nabla\phi_{11} & \cdots & \nabla\phi_{1M} \\ \vdots & \cdots & \vdots \\ \nabla\phi_{N1} & \cdots & \nabla\phi_{NM} \end{bmatrix}^T \begin{bmatrix} \hat{W}_1 \\ \vdots \\ \hat{W}_M \end{bmatrix}^T \tag{49}$$

where $N = 1 + n$, $M = m$, G_i and $\hat{W}_i$ denote the ith element of vector G and $\hat{W}$, respectively, and $\nabla\phi_{ij}$ represents the (i, j)th element of matrix $\nabla\phi$. Substituting $G^T = [\,(1/[B_e(x-x_e)^n])\ \mathbf{O}_n\,]$ into (49) further yields

$$\hat{f} = -\frac{\nabla\phi_{11}\hat{W}_1 + \cdots + \nabla\phi_{m1}\hat{W}_m}{2RB_e(x-x_e)^n}. \tag{50}$$

Since $\nabla\phi_{ij}$ is related to the component of system state X, we can always rewrite (50) into the expression of

$$\hat{f} = (x - x_e)^{-n}(K_1 x - K_2 x_d) \tag{51}$$

where K_1 and K_2 stand for the corresponding gains. This expression describes the relation among the interaction force f, desired position x_d, and current position x. One can see that (51) has a similar form of (17), so the desired impedance model is obtained.

Different from the typical impedance model which has the implicit condition that the environmental equilibrium position retains zero, (51) includes the equilibrium position of the environment. This kind of expression reflects that it emphasizes the relation between the motion penetration and the consequent reaction force, which is more general for contact operations. Besides, this relation is nonlinear, which is in accordance with the nonlinear environmental model. Hence, (51) can be regarded as a variant of the traditional impedance model.

Algorithm 1 Position-Based Implementation for Target Impedance

Input: maximum iteration $N_{\max}$, precision requirement ϑ, downhill factor ψ, loop sentence $\Im$.
Output: reference position x_r.

Initialize $x_r(0) \leftarrow x(0)$, $\psi \leftarrow 1$
Calculate $\mathbb{S}(x_r(0))$, $\nabla\mathbb{S}(x_r(0))$
Calculate $x_r(1) \leftarrow x_r(0) - \psi\frac{\mathbb{S}(x_r(0))}{\nabla\mathbb{S}(x_r(0))}$
Calculate $\mathbb{S}(x_r(1))$, $\nabla\mathbb{S}(x_r(1))$
for $i \leftarrow 1$ to $N_{\max}$ **do**
  $\Im \leftarrow$ **True**
  while $\Im$
    if $|\mathbb{S}(x_r(i))| < |\mathbb{S}(x_r(i-1))|$
      $\Im \leftarrow$ **False**
    else
      $\psi \leftarrow \psi/2$
      $x_r(i) \leftarrow x_r(i-1) - \psi\frac{\mathbb{S}(x_r(i-1))}{\nabla\mathbb{S}(x_r(i-1))}$
    end if
    Calculate $\mathbb{S}(x_r(i))$, $\nabla\mathbb{S}(x_r(i))$
  end while
  if $|x_r(i) - x_r(i-1)| < \vartheta$
    break
  end if
end for

According to (51), we can obtain the reference position x_r since f, x_e, and x_d are available.

Let

$$\mathbb{S}(x) = \hat{f} + \frac{\nabla\phi_{11}\hat{W}_1 + \cdots + \nabla\phi_{m1}\hat{W}_m}{2RB_e(x-x_e)^n} = 0. \tag{52}$$

To solve (52), we employ the Newton Downhill method [43]. The detailed procedure is shown in Algorithm 1. Upon getting x_r, we use it as the position command to control the manipulator, which can realize optimal interaction with the soft environment.

IV. Simulation Studies

In this section, simulation studies are given to verify the effectiveness of the proposed strategy under model uncertainties. The end effector of the manipulator is commanded to move along the X-axis and interact with the environment. The command generator is designed as $r(0) = 0.08$, $\Gamma = -1$, $\alpha = 0.03$, and $\Xi = 1$. Only the interaction force in the X direction is considered and regulated using the proposed optimal impedance control. The environmental parameters are set based on an actual soft object: $K_e = 1149$ N/m, $B_e = 567$ N/m, $n = 1.2$, and $x_e = 0.01$ m. Assume the model uncertainty is $\varepsilon = 1000e\sin(r)$. For the NN setting, the activation function is selected as $[X_1^2\ X_1X_2\ X_2^2]^T$. The initial NN weights are chosen as $\hat{W}(0) = [20\ 20\ 20]^T$, and the learning factor is $\kappa = 5$. To meet the PE condition, we add an exploration noise $\aleph$ to the control input during the adaptive learning stage, where $\aleph(t) = \cos(2.4t)\sin^3(2.4t) + \sin^2(1.12t) + \sin^5(t) + \sin^2(t)\cos(t) + \sin^2(1.2t)\cos(0.5t) + \sin^2(2t)\cos(0.1t)$. In the simulation, the position control is under an ideal case: $x \equiv x_r$,

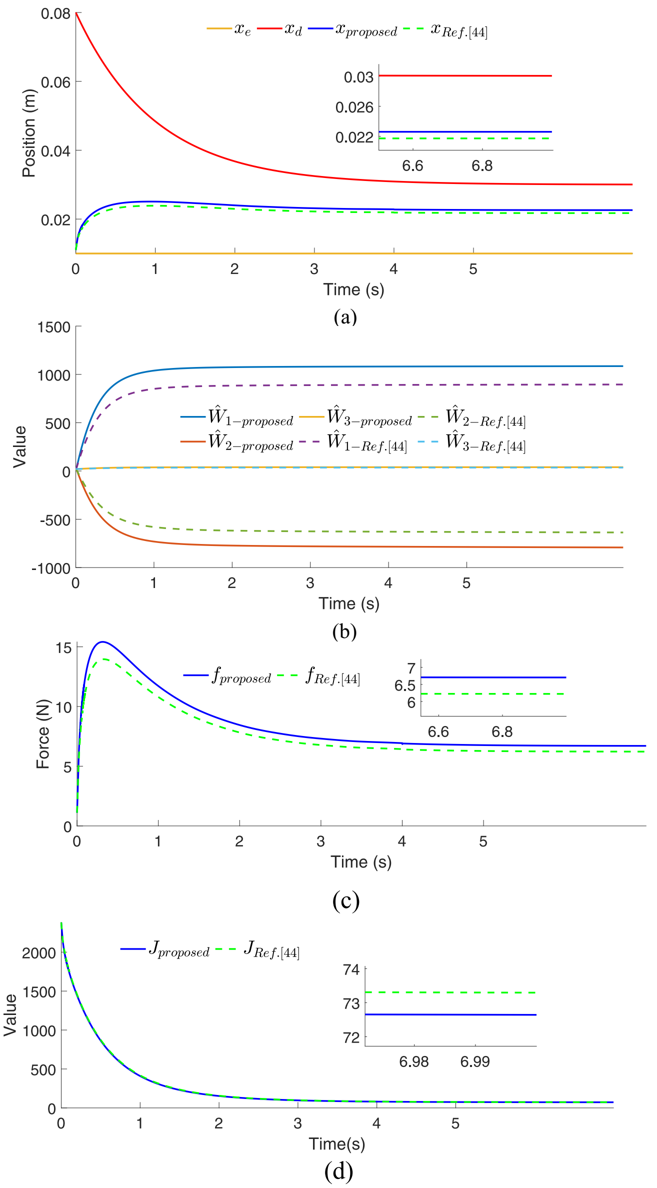

Fig. 2. Interaction under model uncertainties ($Q = 500\,000$, $R = 1$). (a) Evolution of environmental equilibrium position, desired position, and actual position. (b) Convergence of the NN weights. (c) Interaction force. (d) Cost function.

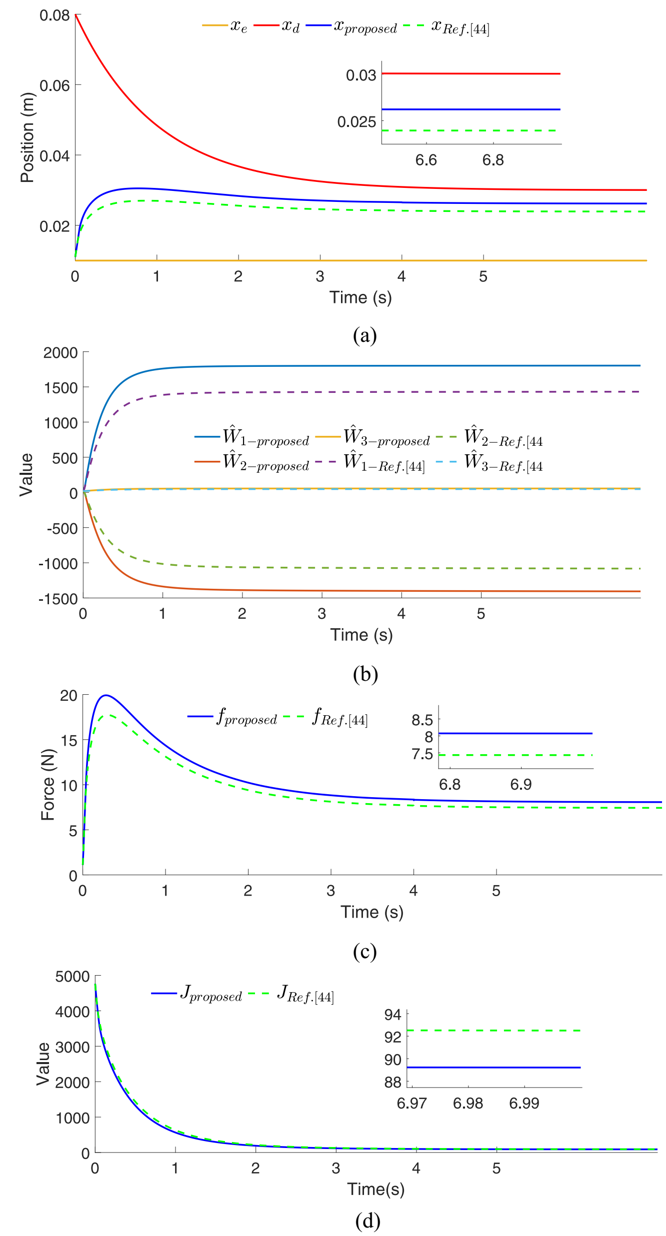

Fig. 3. Interaction under model uncertainties ($Q = 1\,000\,000$, $R = 1$). (a) Evolution of environmental equilibrium position, desired position, and actual position. (b) Convergence of the NN weights. (c) Interaction force. (d) Cost function.

and the sampling time is 0.001 s. The initial position of the end effector is given by $x(0) = 0.011$, and the environmental position is $x_e = 0.01$. For comparison, we also introduce the control method developed by Ref. [44] to learn the optimal impedance model for interaction. Two cases are conducted. In the first case, the weight indexes are set as $Q = 500\,000$, $R = 1$. In the second case, the weights are $Q = 1\,000\,000$, $R = 1$.

Simulation results are presented in Figs. 2 and 3. Figs. 2(a) and 3(a) illustrate the complete trajectories of the interaction process. The final position of the end effector is stabilized between the environmental position and the desired position. A larger value of the performance index Q is expected to result in a smaller position tracking error, which is why the end effector exhibits larger penetration with the environment in the second case leading to a larger interaction force as shown in Figs. 2(c) and 3(c). Figs. 2(b) and 3(b) display the updating process of the NN weights. Figs. 2(d) and 3(d) depict the variation of the cost value. To better compare the interaction performance, the costs for the interaction are summarized in Table II. One can see that both methods successfully reduce the cost to a stable value. The proposed method explicitly accounting for uncertainties in the optimization process using a modified cost function performs superior, which achieves smaller convergent cost and total cost in both cases. The impedance model obtained by solving the standard cost function also performs acceptable. This is because of the inherent ability of the NN-based controller to resist certain

TABLE II
COMPARISON OF COSTS FOR INTERACTION USING THE PROPOSED METHOD AND REF. [44]

	$Q = 500000,\ R = 1$		$Q = 1000000,\ R = 1$	
Quadratic Cost	**proposed**	**Ref. [44]**	**proposed**	**Ref. [44]**
Convergent Cost	72.64	73.30	89.21	92.50
Total Cost	1.67×10^6	1.70×10^6	2.48×10^6	2.66×10^6

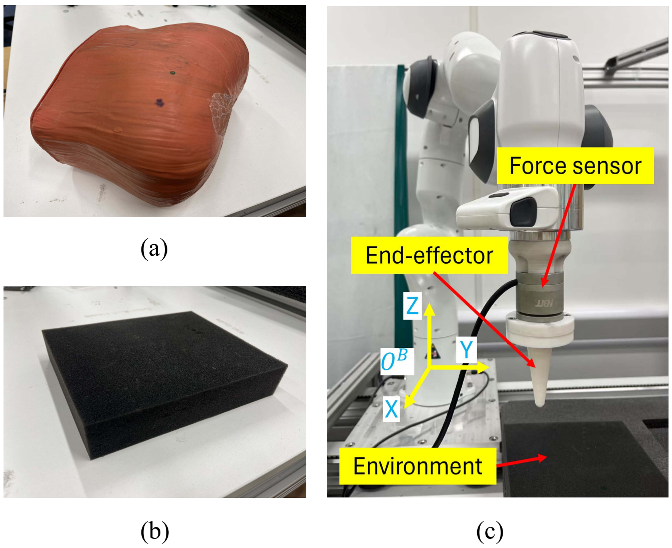

Fig. 4. Experimental setup. (a) Silicone foam. (b) Sponge. (c) Franka robot.

disturbances. It is proven that explicitly handling the model uncertainties using the proposed method can further improve the interaction performance.

V. EXPERIMENTAL STUDIES

To support the proposed strategy for practical interaction, we conduct experiment studies on two different environments: 1) silicone foam and 2) sponge, as shown in Fig. 4(a) and (b). The silicone foam used in the experiments has moderate stiffness and high resilience, with a Shore A hardness of approximately 40. In comparison, the sponge is softer and displays a bit longer restitution time due to its viscoelastic properties, making it slightly slower to recover after deformation. Under identical penetration conditions, the silicone foam generates a larger reaction force than the sponge. The employed robot platform is a seven-degree-of-freedom Franka robot, whose end effector is designed by the user for point interaction with objects. A force sensor is attached to the end effector to collect the real-time contact force. Throughout the process, we control the end effector of the manipulator to move along the Z-axis of the robot base frame O^B. To highlight the interaction performance, we focus only on the contact force that occurs in the Z direction. The whole experimental setup is given in Fig. 4(c). For comparison, we also develop the soft-environment interaction using the strategy proposed by Ref. [45], where the environment was described by the KV model.

TABLE III
ENVIRONMENTAL PARAMETERS IN THE HC MODEL AND THE KV MODEL

	HC			KV	
Environment	K_e**(N/m)**	B_e**(Ns/m)**	n	G_e**(N/m)**	C_e**(Ns/m)**
Silicone Foam	1095	585	1.2	750	20
Sponge	1100	550	2	300	40

TABLE IV
COMPARISON OF FORCE ERRORS USING THE HC MODEL AND THE KV MODEL

	Silicone foam			Sponge		
Model	**ME**	**AE**	**MaxE**	**ME**	**AE**	**MaxE**
HC	0.005	0.13	0.57	-0.18	0.20	0.74
KV	-2.12	2.26	4.69	-1.58	2.19	4.56

A. Environment Identification

We command the end effector of the manipulator to interact with the environment following an excitation trajectory and collect the movement data. The excitation trajectory is specified by

$$x_c(t) = 0.9 - 0.025 * \big[2\sin(0.01\pi t) + 0.25\sin(1.6\pi t) + 0.1\sin(4.2\pi t) + 0.01\sin(4\pi t)\big]. \tag{53}$$

Employing the EWRLS method, we obtain the approximated environmental parameters in the HC model. The same identification process is applied to the KV model, which is expressed as

$$f = G_e(x - x_e) + C_e\dot{x}. \tag{54}$$

The identification results for the silicone foam and sponge using the HC model and KV model are given in Table III. As we can see, the estimated damping and stiffness parameters in the HC model for these two objects are similar, whereas the sponge has a higher n value of 2, showing its prominently damping characteristic [12]. A consistent conclusion can be seen in the estimation results for the KV parameters. The sponge has relatively lower stiffness and higher damping compared to the silicone foam, suggesting that it possesses weaker elastic properties than the silicone foam.

To verify the effectiveness of these two representations for soft environments, we estimate the interaction forces based on the identified models, respectively, and compare them with the measurements from the force sensor. The results are displayed in Fig. 5. One can see that the force derived from the HC model closely aligns with the measurements of the force sensor. In contrast, the force from the KV model exhibits a substantial deviation from the sensor readings, especially during the recovery phase. To better evaluate the accuracy of the model descriptions, we introduce three evaluation metrics: 1) mean error (ME); 2) absolute error (AE); and 3) maximum error (MaxE). The results for the force estimation error are summarized in Table IV. It is obvious that the HC model consistently outperforms the KV model in all metrics in terms of force estimation for soft environments, making it a more reliable choice for precise contact control applications.

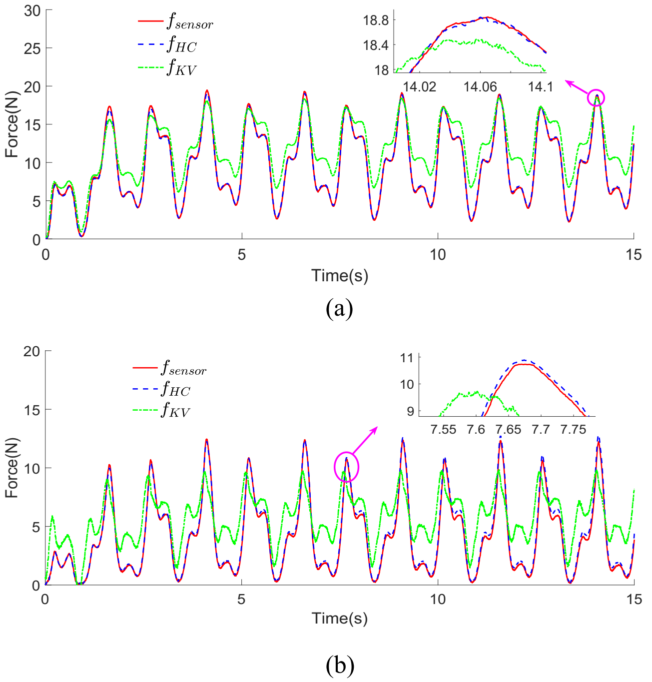

Fig. 5. Comparison of estimated force in HC and KV models. (a) Silicone foam. (b) Sponge.

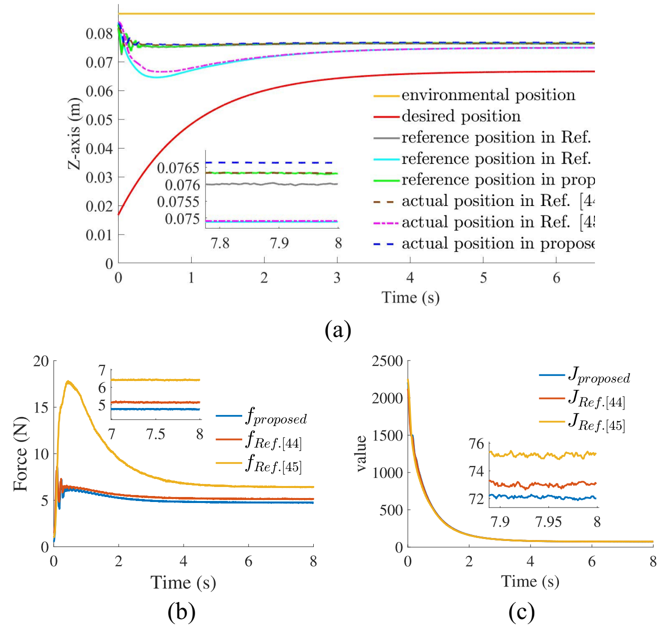

Fig. 6. Interaction with silicone foam ($Q = 500\,000$, $R = 1$). (a) Evolution of the environmental equilibrium position, desired position, reference position, and actual position. (b) Interaction force. (c) Cost function.

B. Optimal Interaction With Soft Environments

In this section, we will further investigate the interaction performance of the proposed method in practical operation. For comparative analysis, we consider two baseline controllers: one employs the nonlinear optimal control method proposed in Ref. [44], which does not account for model uncertainties, and the other is based on the KV model from Ref. [45]. The control system is established using the identified models presented in Section V-A. In the experiment, we enforce the end effector of the manipulator to interact with silicone foam and sponge, respectively. The end effector of the manipulator initially maintains slight contact with the object. Within the interaction process, the velocity of the end effector is limited to $\dot{x} \leq (0.1K_e/B_e)$ to meet the conditions of the HC model.

The first scenario is to interact with the silicone foam. The environmental equilibrium position is identified using the 3-D point cloud technique, which is obtained as $x_e = 0.08669$. We design the command generator: $r(0) = 0.01669$, $\Gamma = -1$, $\alpha = -0.05$, and $\Xi = 1$. The exploration noise is designed as $\aleph(t) = \sum_{i=1}^{10} (0.1/i)\sin(k * i)$. The activation function is selected as $[X_1^2\ X_1X_2\ X_2^2]^T$. Two case studies with different weight settings of the cost function are considered. In the first case, $Q = 500\,000$, $R = 1$. The NN is initialized as $\kappa = 5$ and $\hat{W}(0) = [\,30\ 30\ 30\,]^T$. In the second case, $Q = 1500\,000$, $R = 1$, $\kappa = 5$, and $\hat{W}(0) = [\,250\ 250\ 250\,]^T$.

The results are presented in Figs. 6 and 7, where Figs. 6(a) and 7(a) illustrate the actual position of the end effector during interaction, Figs. 6(b) and 7(b) depict the interaction force, and Figs. 6(c) and 7(c) show the associated cost functions. As observed, the reference and actual positions exhibit oscillations during the interaction regulation phase due to the online adaptation process, which quickly decay as the parameters converge. In both cases, the proposed method achieves a lower convergent cost, indicating that the obtained impedance model is more effective in ensuring the desired interaction behavior. The control performance of the proposed method is more pronounced in the second case, where the weight for the tracking error is larger. This leads to greater penetration into the environment, resulting in larger reaction forces. However, the compared method based on the KV model struggles to handle large penetrations due to its limited force-estimation capacity as the result shown in Fig. 5(a), leading to poorer interaction performance. Similar results are observed for the method from Ref. [44], as it does not adequately address model uncertainties. To provide a more quantitative comparison, we summarize the costs in Table V, which shows that both the convergent cost and total cost are consistently lower using the proposed method. Specifically, the proposed method achieves a 1.42% cost reduction compared to Ref. [44] and 4.25% compared to Ref. [45] in the first case. In the second case, it achieves a 4.55% reduction compared to Ref. [44] and 8.82% compared to Ref. [45], demonstrating its superior interaction performance and cost efficiency.

The second group of experiments focuses on evaluating the interaction performance with the sponge. The command generator is configured with parameters set to $r(0) = 0.025$, $\Gamma = 1$, $\alpha = 0.02$, and $\Xi = 1$. The environmental equilibrium position is identified as $x_e = 0.075$. The exploration noise and the activation function remain consistent with previous settings. Similarly, we design two case studies. In the first case, the cost weights are given by $Q = 100\,000$ and $R = 1$. The NN is initialed with weights $\hat{W}(0) = [\,10\ 10\ 10\,]^T$ and learning rate $\kappa = 1$. In the second case, the cost weights are

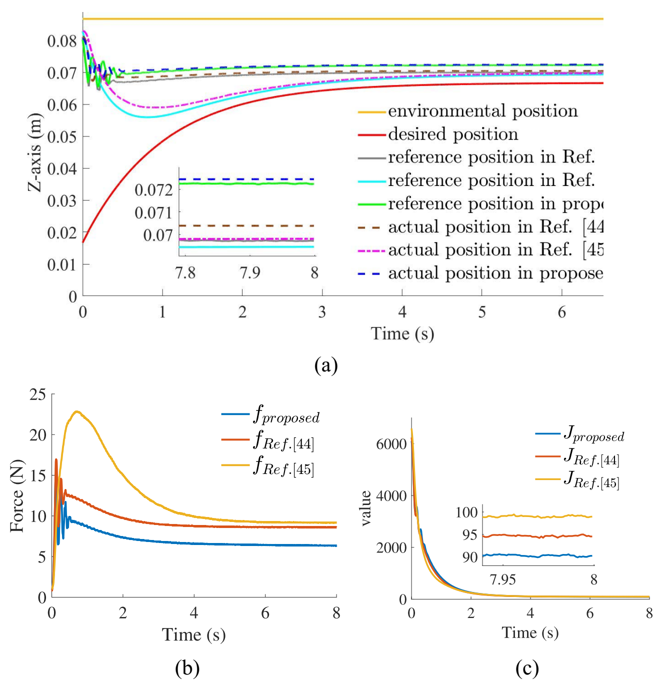

Fig. 7. Interaction with silicone foam ($Q = 1500\,000$, $R = 1$). (a) Evolution of the environmental equilibrium position, desired position, reference position, and actual position. (b) Interaction force. (c) Cost function.

TABLE V
COSTS OF THE INTERACTION WITH SILICONE FOAM

	$Q = 500000$, $R = 1$		
Quadratic Cost	**proposed**	**Ref. [44]**	**Ref. [45]**
Convergent Cost	72.02	73.06	75.22
Total Cost	1.68×10^6	1.71×10^6	1.72×10^6
	$Q = 1500000$, $R = 1$		
Quadratic Cost	**proposed**	**Ref. [44]**	**Ref. [45]**
Convergent Cost	90.24	94.54	98.97
Total Cost	3.18×10^6	3.21×10^6	3.36×10^6

TABLE VI
COSTS OF THE INTERACTION WITH SPONGE

	$Q = 100000$, $R = 1$		
Quadratic Cost	**proposed**	**Ref. [44]**	**Ref. [45]**
Convergent Cost	23.84	24.59	29.78
Total Cost	2.73×10^5	2.88×10^5	3.38×10^5
	$Q = 1000000$, $R = 1$		
Quadratic Cost	**proposed**	**Ref. [44]**	**Ref. [45]**
Convergent Cost	55.51	61.76	78.90
Total Cost	9.27×10^5	1.03×10^6	1.30×10^6

$Q = 1000\,000$, $R = 1$. The NN is designed as initial weights $\hat{W}(0) = [\,60\ 60\ 60\,]^T$ and the learning rate $\kappa = 5$.

Figs. 8(a) and 9(a) illustrate the trajectory evolution. Fig. 8(b) and (b) display the interaction force. The variation of costs is depicted in Figs. 8(c) and 9(c). Under the control of the proposed method, the end effector is adjusted to a position closer to the environmental surface compared to those in

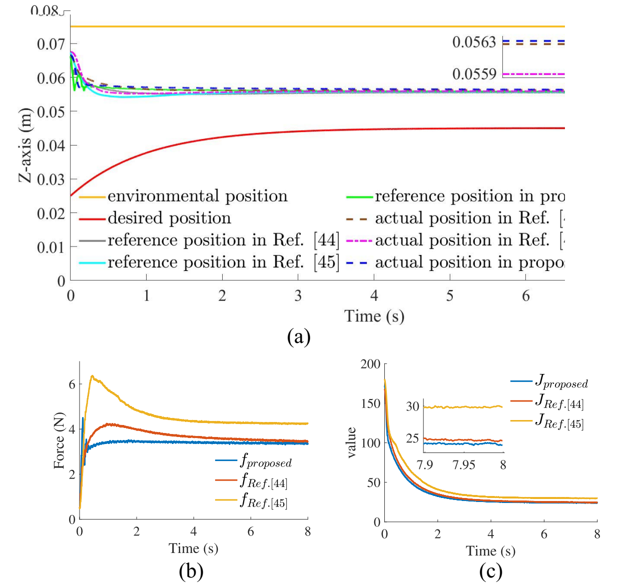

Fig. 8. Interaction with sponge ($Q = 100\,000$, $R = 1$). (a) Evolution of the environmental equilibrium position, desired position, reference position, and actual position. (b) Interaction force. (c) Cost function.

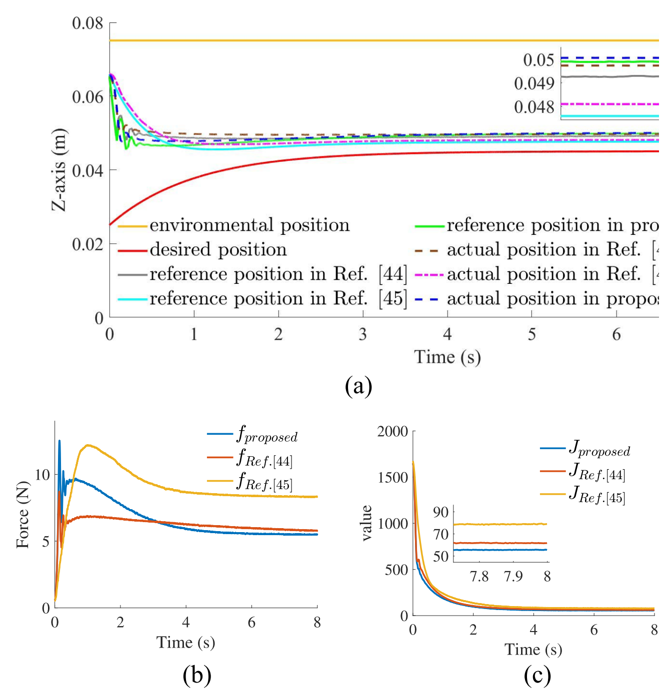

Fig. 9. Interaction with sponge ($Q = 1\,000\,000$, $R = 1$). (a) Evolution of the environmental equilibrium position, desired position, reference position, and actual position. (b) Interaction force. (c) Cost function.

Ref. [44] and Ref. [45], achieving lower-cost performance. A summary of the quadratic costs for the interaction is provided in Table VI, revealing that both the convergent cost and total cost in the proposed method are significantly smaller than those of the compared methods. Specifically, compared to Ref. [44], the proposed method reduces the convergent cost by 3.05% in the first case and by 10.12% in the second case. When compared to Ref. [45], it achieves a 19.95% reduction in the first case and a 29.65% reduction in the second case. The improvement is particularly evident when compared to the first group of experiments, primarily due to the viscoelastic properties of the sponge. The presence of significant model uncertainties and the inability of the KV model to sufficiently represent the soft environment results in inadequate impedance models and ultimately lead to degraded interaction performance.

VI. Conclusion

In this article, we propose a novel NN-based optimal impedance control strategy for robots to achieve optimal interaction with soft environments. In regard to the nonlinear properties of soft environments, we develop an HC model to describe the environmental dynamics. Then, the environmental parameters are identified using the collected interactive data. To achieve accurate control, the unknown model uncertainties are also considered in the design of the impedance controller, wherein we introduce an auxiliary system and a modified cost function. To avoid mathematically solving the nonlinear HJB equation of the optimal control problem, NN is employed to adaptively learn the approximate HJB solution. Finally, the desired impedance model is obtained and realized via the position-based implementation, where the reference position is acquired using the Newton Downhill method. In the future, we will focus on studying the optimal interaction control without prior identification of the environmental dynamics, which will offer greater flexibility for practical contact operations. In addition, more complex environments would be considered, specifically, those with time-varying parameters or subject to abrupt changes. These cases present significant challenges for traditional control models. Our goal is to develop adaptive strategies that can handle such complexities effectively.

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